

Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar

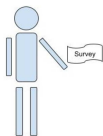
September 2026

The basic problem

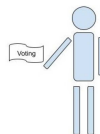
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

The basic problem

We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

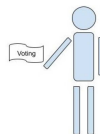
We want the average response in a target population, in which we observe only covariates.

The problem is that the populations may be very different, maybe leading to bias.



Photo copyright Mark Taylor: natereport.com

Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

The basic problem

We have a survey population, for whom we observe:

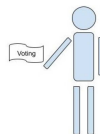
- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.

The problem is that the populations may be very different, maybe leading to bias.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[y|x_j]$.

The basic problem

We have a survey population, for whom we observe:

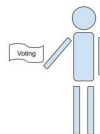
- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.

The problem is that the populations may be very different, maybe leading to bias.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[y|x_j]$.

Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

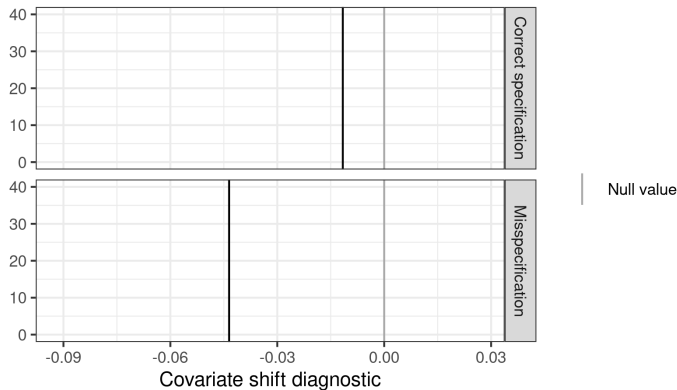


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

Result preview

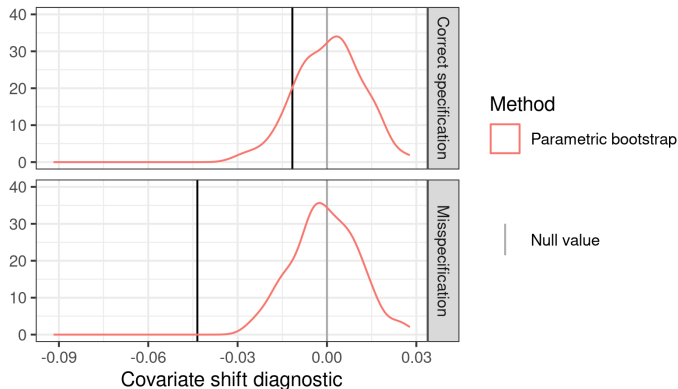


Figure 1: A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

Result preview

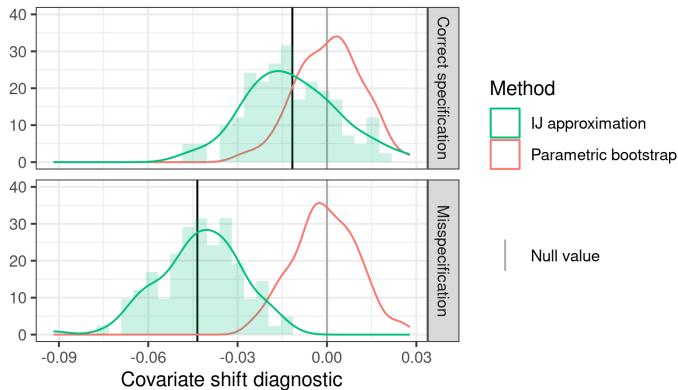


Figure 1: A covariate shift diagnostic

IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

Result preview

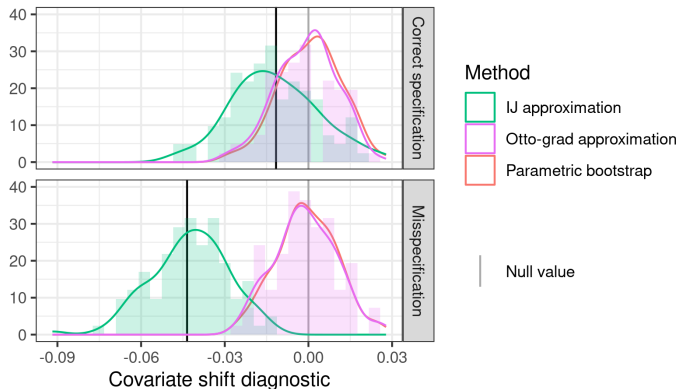


Figure 1: A covariate shift diagnostic

IJ confidence intervals are fast and often work well for posterior means. But they are not accurate for this more complex diagnostic (high Monte Carlo error).

Answer: Our flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

- Bayesian local regression specification checks
 - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
 - (With Lucas Schwengber)

TODO: put in pictures

Bayesian local regression specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [Di Noia et al., Cabral et al.]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
 - The task is not to predict on a new IID survey datapoint

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [Di Noia et al., Cabral et al.]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
 - The task is not to predict on a new IID survey datapoint

The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\substack{\text{Want to check specification} \\ \text{without re-running MCMC}}}.$$

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\substack{\text{Want to check specification} \\ \text{without re-running MCMC}}}.$$

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$
This is what we actually want — let's check this directly.

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

$$\text{Correct specification} \quad \Rightarrow \quad \underbrace{\text{Regression estimates are invariant to } \mathcal{P}_S(x)}_{\text{This is what we actually want — let's check this directly.}}$$

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\substack{\text{Want to check specification} \\ \text{without re-running MCMC}}}.$$

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

$$\text{Correct specification} \quad \Rightarrow \quad \underbrace{\text{Regression estimates are invariant to } \mathcal{P}_S(x)}_{\text{This is what we actually want — let's check this directly.}}$$

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Lots of similar ideas in causal inference, kernel learning, transfer learning [Bühlmann, Rojas-Carulla et al., Muandet et al.]. This is just a local Bayesian diagnostics version.

Simple visual example

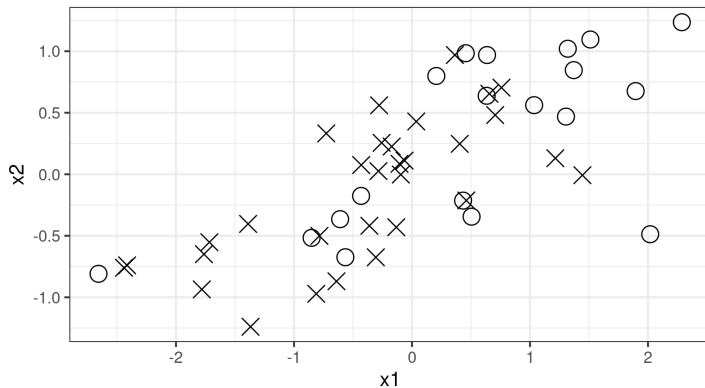


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

Simple visual example

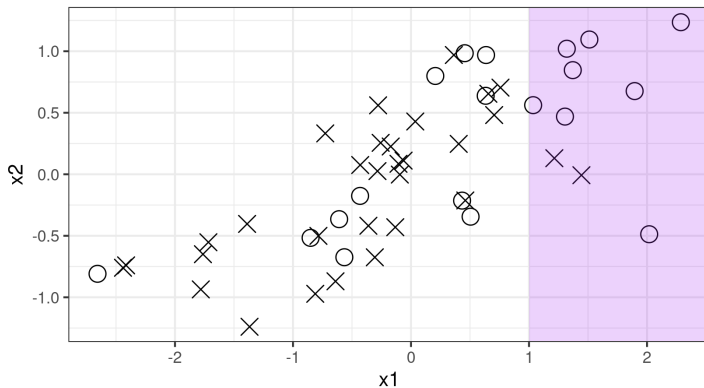


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

Taking $w(x) = \mathbb{I}(x_1 > 1)$, we upweight positive examples.

Simple visual example

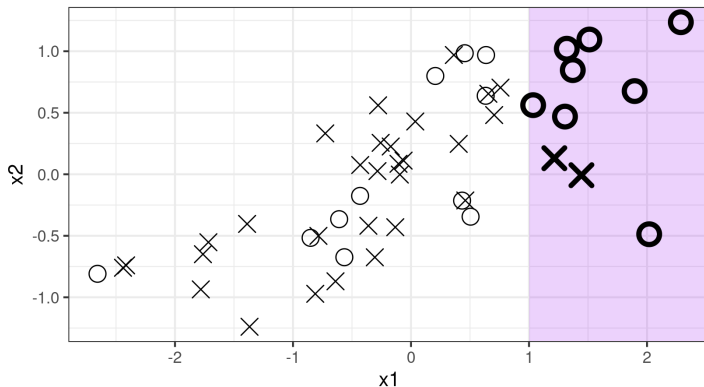


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

Taking $w(x) = \mathbb{I}(x_1 > 1)$, we upweight positive examples.

With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

Theorem (Diaconis and Freedman, Gustafson [1996], Giordano et al. [2018]):

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

Theorem (Diaconis and Freedman, Gustafson [1996], Giordano et al. [2018]):

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left. \frac{\partial \hat{\mu}}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\underbrace{\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta)}_{\text{Log model perturbation}} \right)$$

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

Bootstrapping Bayesian sensitivity analysis

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

Bootstrapping Bayesian sensitivity analysis

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

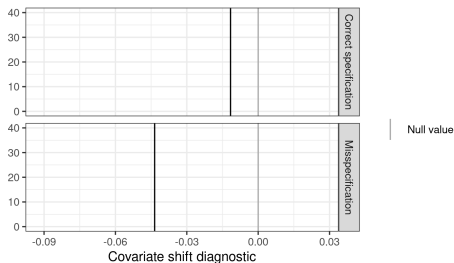


Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

Model 1 (misspecified) : $y \sim \text{Logistic}(1 + X3 + X4 + X5)$

Bootstrapping Bayesian sensitivity analysis

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

To assess “closeness to zero,” compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \dots, 100$:
 1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \dots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ (**time consuming!**)
 3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

Bootstrapping Bayesian sensitivity analysis

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

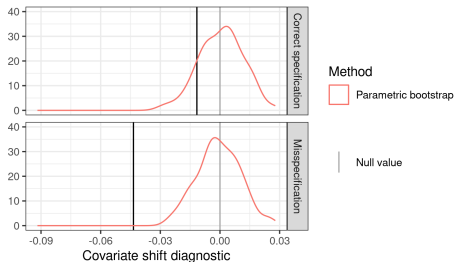


Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

Model 1 (misspecified) : $y \sim \text{Logistic}(1 + X3 + X4 + X5)$

Frequentist variance with the infinitesimal
jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification [Huggins and Miller, 2023, Wang et al., 2026, Luo and Ji, 2026, Ji et al., 2025]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)} [\phi(\theta)] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (1 - 0) \\ &= \end{aligned}$$

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)} [\phi(\theta)] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \bigg|_{\epsilon=0} (1 - 0) \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \sum_{i=1}^{N_S} \underbrace{\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial w_i} \bigg|_{w_i=0}}_{\text{"Empirical influence function"} \frac{\psi_i}{N_S}} w_i \end{aligned}$$

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp \left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta) \right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)} [\phi(\theta)] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (1 - 0) \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \sum_{i=1}^{N_S} \underbrace{\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial w_i} \right|_{w_i=0}}_{\text{"Empirical influence function"} \frac{\psi_i}{N_S}} w_i \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \frac{1}{N_S} \sum_{i=1}^{N_S} \psi_i w_i \end{aligned}$$

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp \left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta) \right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)} [\phi(\theta)] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (1-0) \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \sum_{i=1}^{N_S} \underbrace{\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial w_i} \right|_{w_i=0}}_{\text{"Empirical influence function"} \frac{\psi_i}{N_S}} w_i \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \frac{1}{N_S} \sum_{i=1}^{N_S} \psi_i w_i \end{aligned}$$

If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, we study the *von Mises expansion* [von Mises, 1947].

Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t} \right|_{t=0} (1-0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0, 1]) \\ &= \underbrace{\sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon (\ell(y_i^*|x_i, \theta) - \ell(y_i|x_i, \theta)) \right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon (\ell(y_i^*|x_i, \theta) - \ell(y_i|x_i, \theta)) \right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

Note: The von-Mises expansion is stronger than necessary for consistency [Giordano and Broderick, 2026]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

Data re-weighting.

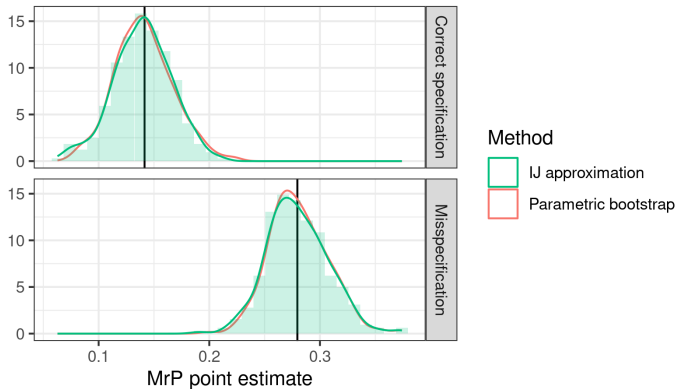


Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [2026], Ji et al. [2025].

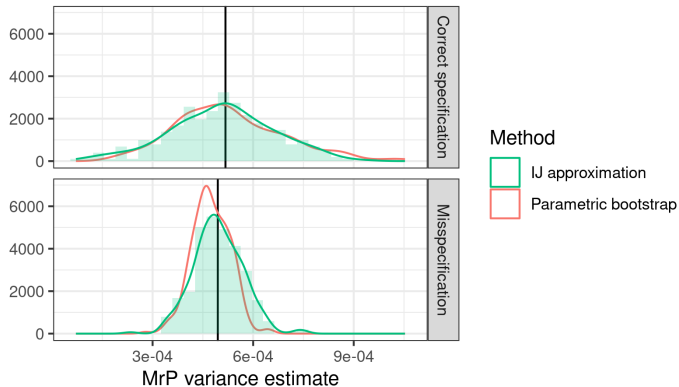


Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

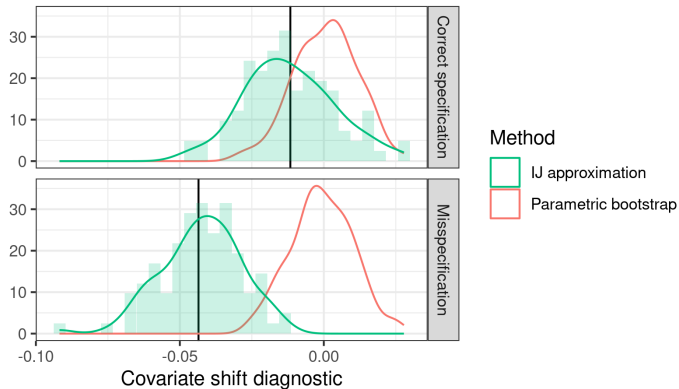


Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Local robustness with flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)} [\phi(\theta)] \approx \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws [Gustafson, 1996, Giordano et al., 2018, Cabral et al., Di Noia et al., Giordano et al., 2026].

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations (CITE)

Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^{\epsilon}$.

There's a large literature on learning flows between distant distributions. (CITE)

Our problem is easier:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

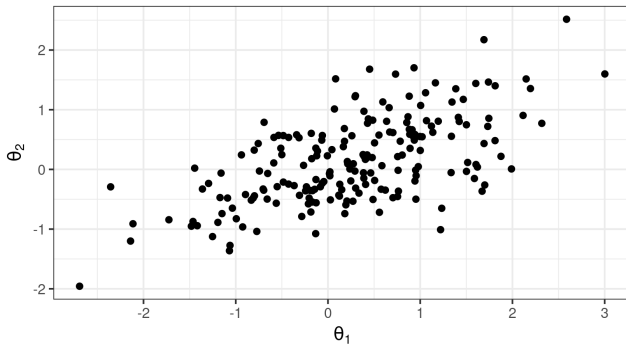


Figure 5: Draws from a posterior

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

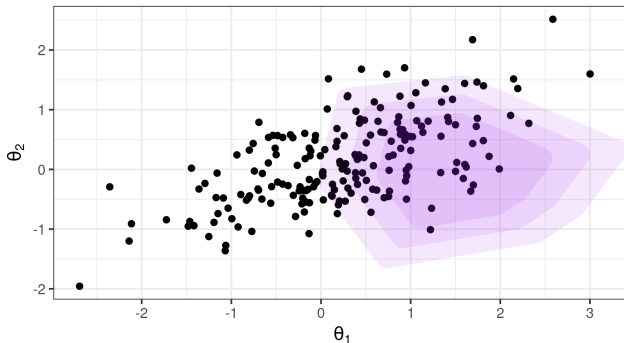


Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined “vertically.”

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

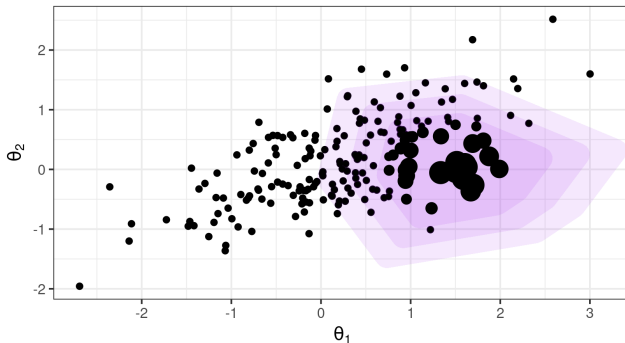


Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting (Appendix B Giordano et al. [2018])

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

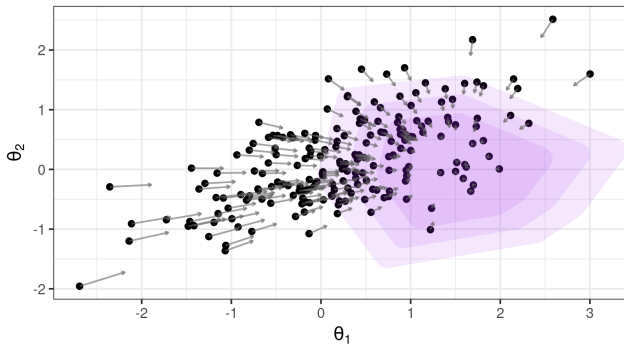


Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto f^\epsilon(\theta)$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \quad \text{(good flow)}$$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \quad \text{(good flow)}$$

$$\left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta + \epsilon \nabla v(\theta))] \Rightarrow \quad \text{(smoothness)}$$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \quad \text{(good flow)}$$

$$\left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta + \epsilon \nabla v(\theta))] \Rightarrow \quad \text{(smoothness)}$$

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi(\theta)^\top \nabla v(\theta)] \quad \text{(Connecting } h \text{ and } v)$$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ , $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \quad (\text{good flow})$$

$$\left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] = \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta + \epsilon \nabla v(\theta))] \Rightarrow \quad (\text{smoothness})$$

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi(\theta)^\top \nabla v(\theta)] \quad (\text{Connecting } h \text{ and } v)$$

Theorem (Schewengber 2026):

Under mild regularity conditions there exists a solution v to the differential equation

$$\mathcal{L}_{\mathcal{P}}(v) := -\nabla v^\top \nabla \log \mathcal{P} - \Delta v = h$$

such that $\|\nabla v\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon \nabla v(\theta)$ satisfies Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of Langevin diffusion, and the trace of the Stein operator.[Ambrosio et al., 2021, Song et al., 2021, Ross, 2011]

How can we find $v(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $v(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla v(\theta)] \quad (\text{Identity connecting } h \text{ and } v)$$

Basis expansion approach (Otto's calculus [Otto, 2001]):

- Specify a family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $v(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

How can we find $v(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $v(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla v(\theta)] \quad (\text{Identity connecting } h \text{ and } v)$$

Basis expansion approach (Otto's calculus [Otto, 2001]):

- Specify a family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $v(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \text{Cov}_{\mathcal{P}(\theta)}(\phi_n(\theta), h(\theta)) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} \vdots & & \\ \dots & \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi_n(\theta)^\top \nabla b_m(\theta)] & \dots \\ \vdots & & \end{bmatrix}}_{N \times M \text{ matrix}} \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

How can we find $v(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $v(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla v(\theta)] \quad (\text{Identity connecting } h \text{ and } v)$$

Basis expansion approach (Otto's calculus [Otto, 2001]):

- Specify a family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $v(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \text{Cov}_{\mathcal{P}(\theta)}(\phi_n(\theta), h(\theta)) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} \vdots & & \\ \dots & \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi_n(\theta)^\top \nabla b_m(\theta)] & \dots \\ \vdots & & \end{bmatrix}}_{N \times M \text{ matrix}} \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

Theorem (Schewengber 2026): When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $v(\theta)$ in $L_2(\mathcal{P})$.

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Using the 1st order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_m$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

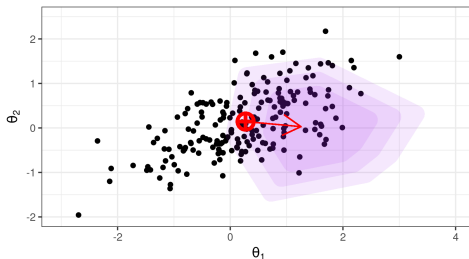
An elegant flow

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Using the 1st order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_n$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!



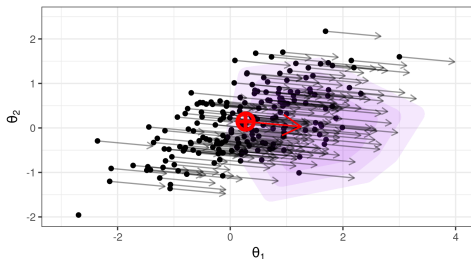
An elegant flow

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Using the 1st order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_n$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!



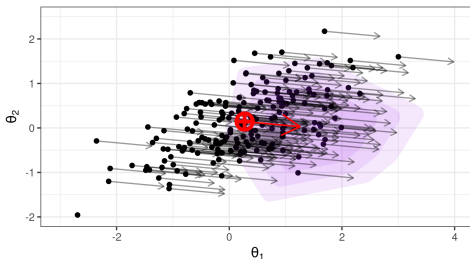
An elegant flow

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Using the 1st order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_n$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!



Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i .

Set $\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

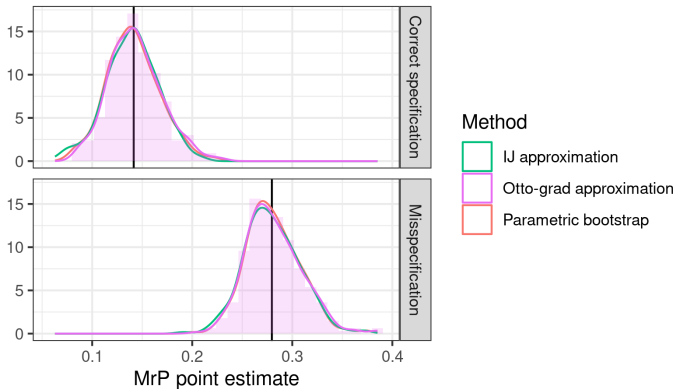


Figure 6: MrP Posterior Mean

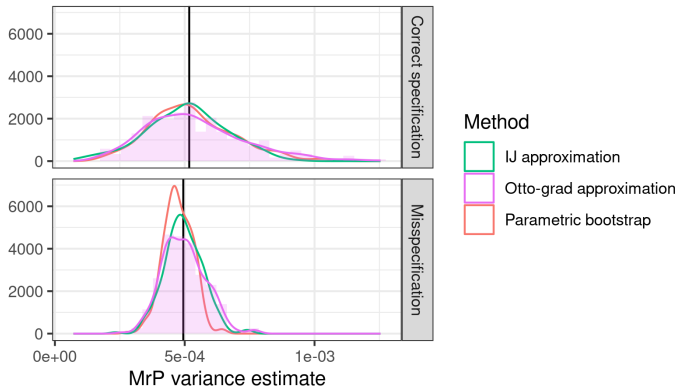


Figure 6: MrP Posterior Variance

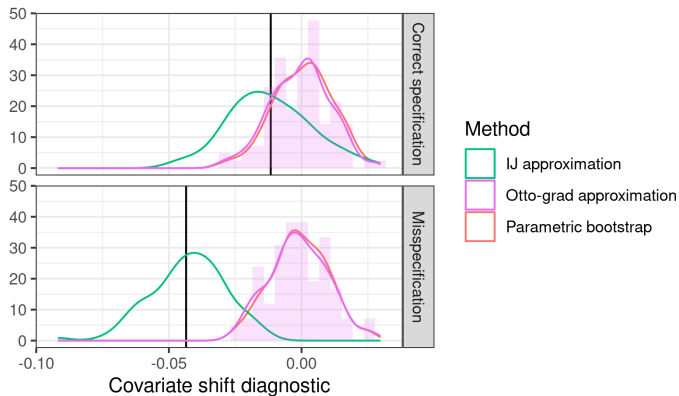


Figure 6: Covariate Shift Diagnostic

Understanding the gap between IJ and Otto is ongoing work!

- Bayesian local regression specification checks
 - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Flow-based Bayesian local robustness: “Otto grad”
 - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

- Luigi Ambrosio, Elia Brué, Daniele Semola, et al. *Lectures on optimal transport*, volume 130. Springer Cham, 2021.
- Peter Bühlmann. Invariance, Causality and Robustness: 2018 Neyman Lecture. 35(3):404–426. ISSN 0883-4237. URL <https://www.jstor.org/stable/26997912>.
- Rafael Cabral, David Bolin, and Håvard Rue. Robustness, model checking, and hierarchical models. 87(3):632–652. ISSN 1369-7412, 1467-9868. doi: 10.1093/jrsssb/qkae107. URL <https://academic.oup.com/jrsssb/article/87/3/632/7914952>.
- Antonio Di Noia, Fabrizio Ruggeri, and Antonietta Mira. Likelihood Distortion and Bayesian Local Robustness. 20(4). ISSN 1936-0975. doi: 10.1214/25-BA1526. URL <https://projecteuclid.org/journals/bayesian-analysis/volume-20/issue-4/Likelihood-Distortion-and-Bayesian-Local-Robustness/10.1214/25-BA1526.full>.
- P. Diaconis and D. Freedman. On the consistency of Bayes estimates. pages 1–26.
- R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.
- R. Giordano, T. Broderick, and M. Jordan. Covariances, robustness and variational Bayes. *The Journal of Machine Learning Research*, 19(1):1981–2029, 2018.
- R. Giordano, A. Cima, J. Murray, E. Hartman, and A. Feller. Locally equivalent weights for multilevel regression and poststratification, 2026. URL <https://arxiv.org/abs/2606.04250>.
- P. Gustafson. Local sensitivity of posterior expectations. *The Annals of Statistics*, 24(1):174–195, 1996.
- J. Huggins and J. Miller. Reproducible model selection using bagged posteriors. *Bayesian Analysis*, 18(1):79–104, 2023.
- Feng Ji, JoonHo Lee, and Sophia Rabe-Hesketh. Valid Standard Errors for Bayesian Quantile Regression With Clustered and Independent Data. page 10769986251379738, 2025. ISSN 1076-9986. doi: 10.3102/10769986251379738. URL <https://doi.org/10.3102/10769986251379738>.
- Swen Kuh, Lauren Kennedy, Qixuan Chen, and Andrew Gelman. Using leave-one-out cross validation (LOO) in a multilevel regression and poststratification (MRP) workflow: A cautionary tale. 43(5):953–982. ISSN 1097-0258. doi: 10.1002/sim.9964. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.9964>.
- Nanyu Luo and Feng Ji. Robust Standard Errors for Bayesian Posterior Functionals via the Infinitesimal Jackknife, 2026. URL <http://arxiv.org/abs/2604.03398>.
- Krikamol Muandet, Wittawat Jitkrittum, and Jonas Kübler. Kernel Conditional Moment Test via Maximum Moment Restriction. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 41–50. PMLR. URL <https://proceedings.mlr.press/v124/muandet20a.html>.
- Felix Otto. The Geometry of Dissipative Evolution Equations: The Porous Medium Equation. *Communications in Partial Differential Equations*, 26(12):1491–1574, 2001. ISSN 0360-5302. doi: 10.1081/PDE.100002242. URL <https://doi.org/10.1081/PDE.100002242>.